Benchmarking Superconducting Qubits with Randomized Benchmarking in Qiskit

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***Abstract*— This project investigates the fidelity of quantum gates in superconducting qubits using Randomized Benchmarking (RB) and Interleaved Randomized Benchmarking (IRB). Experiments will be implemented in Qiskit and executed both on noisy simulators with device-calibrated noise models and on real IBM Quantum hardware. The study aims to extract the error per Clifford (EPC) for single- and two-qubit gates, analyse discrepancies between simulation and hardware, and relate these errors to physical noise mechanisms such as decoherence times (T_1, T_2) and readout errors. Results will provide quantitative insight into hardware limitations and contribute to the understanding of gate performance in near-term quantum devices, while demonstrating practical implementation of quantum benchmarking protocols.**

I. Introduction

THIS Superconducting qubits are among the leading platforms for near-term quantum computing, offering fast gate operations and scalability potential. However, their performance is limited by decoherence, control errors, and readout imperfections. Quantifying the fidelity of quantum gates is therefore essential to understand hardware limitations and to guide optimisation strategies.

Randomized Benchmarking (RB) is a widely used protocol to measure the average error per Clifford gate in a scalable and robust manner, mitigating the effects of state preparation and measurement errors. Interleaved RB (IRB) extends this approach to estimate the fidelity of specific gates, such as CNOT, which are critical for universal quantum computation.

In this project, we implement RB and IRB using Qiskit and Qiskit Experiments, executing experiments both in simulators with noise models extracted from IBM Quantum devices and on real superconducting hardware. This approach allows us to compare idealised noise models with actual hardware performance, relate errors to physical mechanisms such as T_1, T_2, and readout noise, and provide insights into the reliability of quantum operations in near-term devices.

II. Proposition of methods

The project will follow a structured approach combining simulation and hardware experiments:

1. Simulation with Noise Models: Using Qiskit Aer, we will simulate standard and interleaved Randomized Benchmarking (RB/IRB) sequences with noise models derived from IBM Quantum backends. This will provide baseline estimates of gate fidelities and expected decay curves.

2. Hardware Experiments: RB and IRB sequences will be executed on an accessible IBM Quantum superconducting processor, respecting the device topology and calibration parameters. Single- and two-qubit gates, including CNOT, will be characterised.

3. Data Analysis: Survival probabilities will be fitted to exponential decays to extract the Error Per Clifford (EPC) and gate-specific fidelities. Differences between simulated and hardware results will be analysed and correlated with physical parameters such as T_1, T_2, and readout error.

4. Comparison with Literature: Results will be contextualised with previous experimental benchmarks and state-of-the-art fidelities for superconducting qubits, highlighting limitations and potential improvements in control strategies.

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III. Expected Outcomes

This project is expected to deliver quantitative estimates of gate fidelities for single- and two-qubit operations on superconducting hardware, validated against noise-model simulations. Differences between simulator and real-device results will be analysed and related to physical error sources such as decoherence and readout errors. The work will provide insight into hardware limitations, demonstrate practical implementation of quantum benchmarking in Qiskit, and offer guidance for improving gate performance in near-term quantum processors.

References

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